



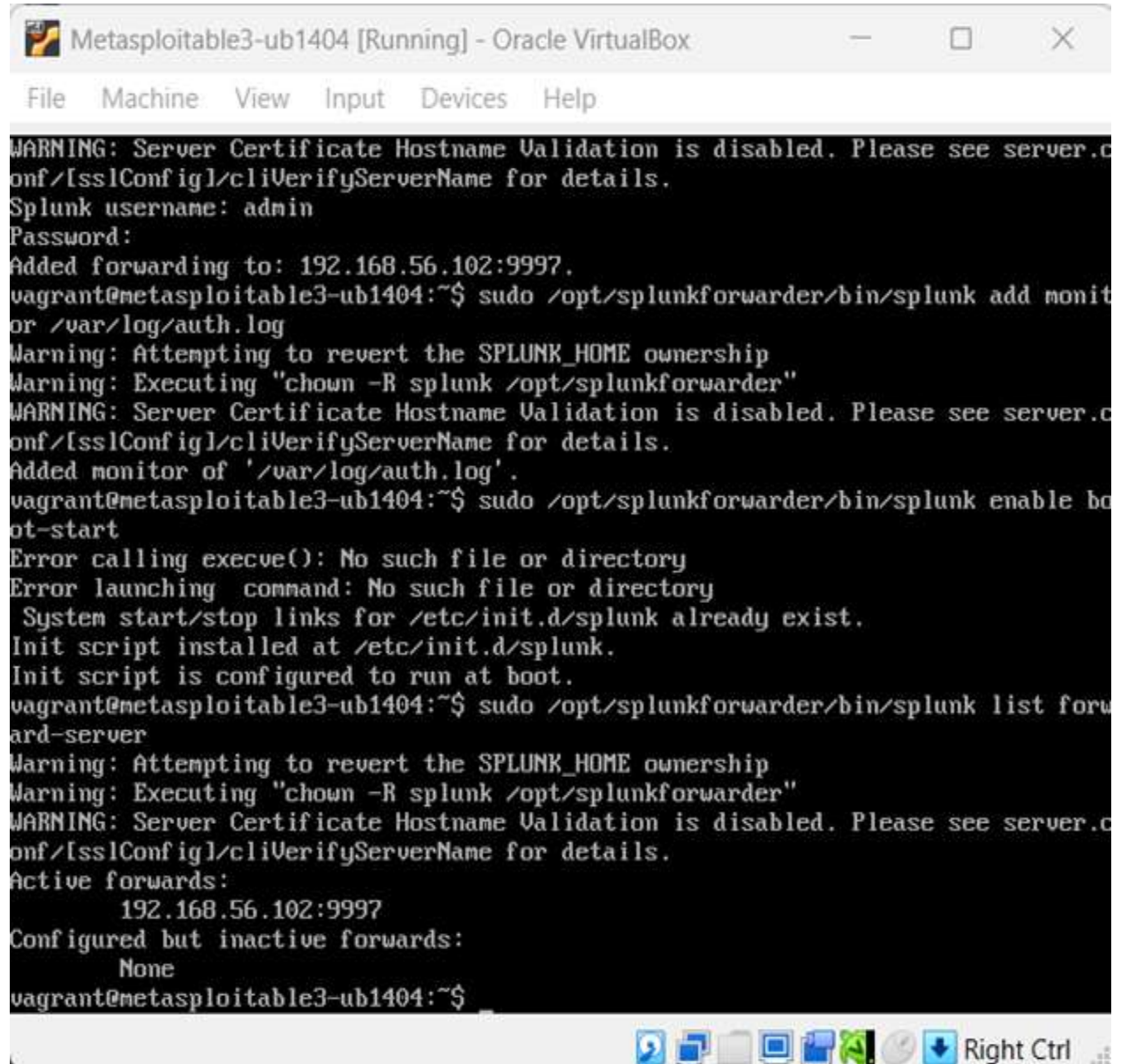
ICS344PROJECT: PHASE 1

Section:1 Group:9



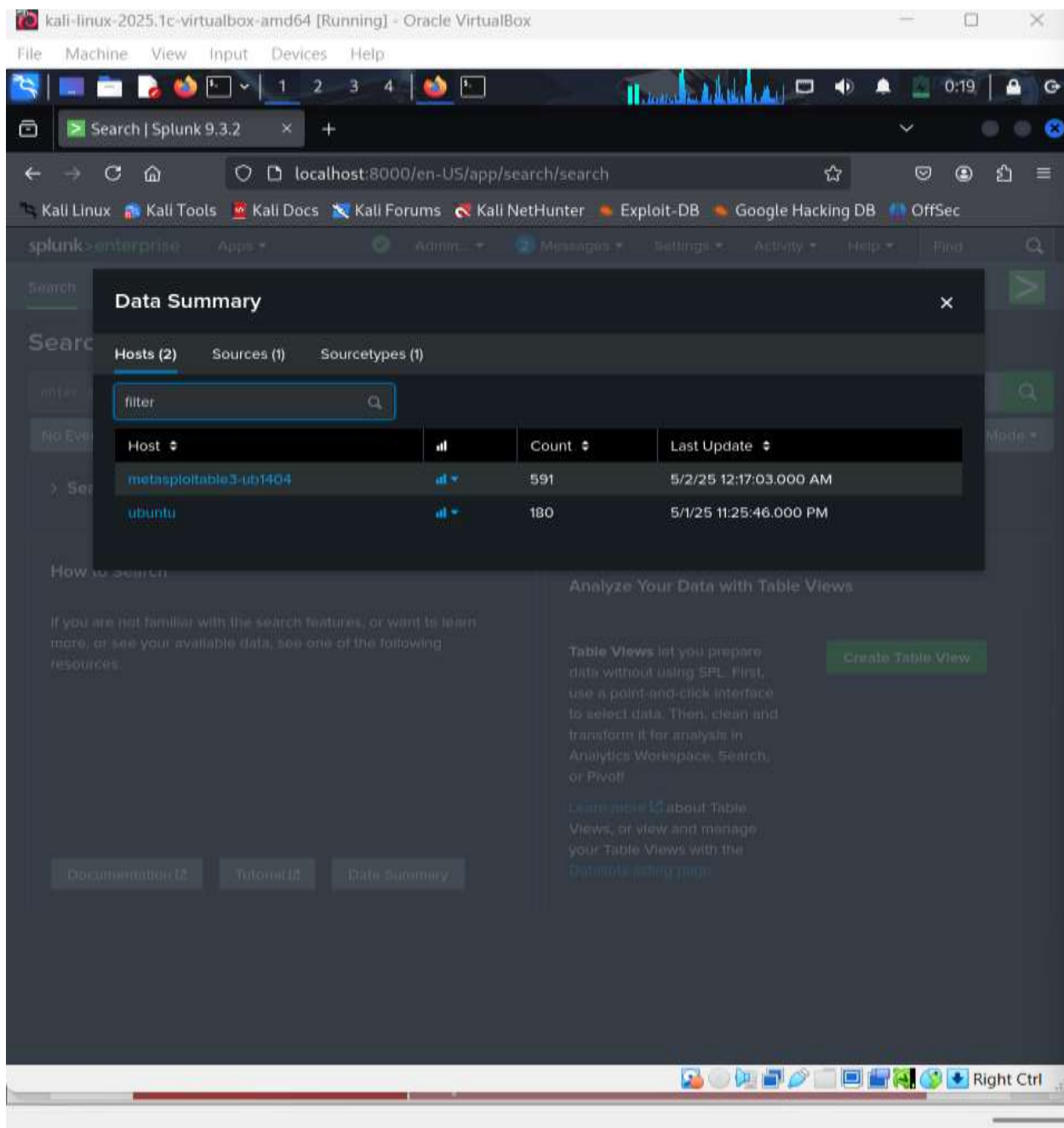
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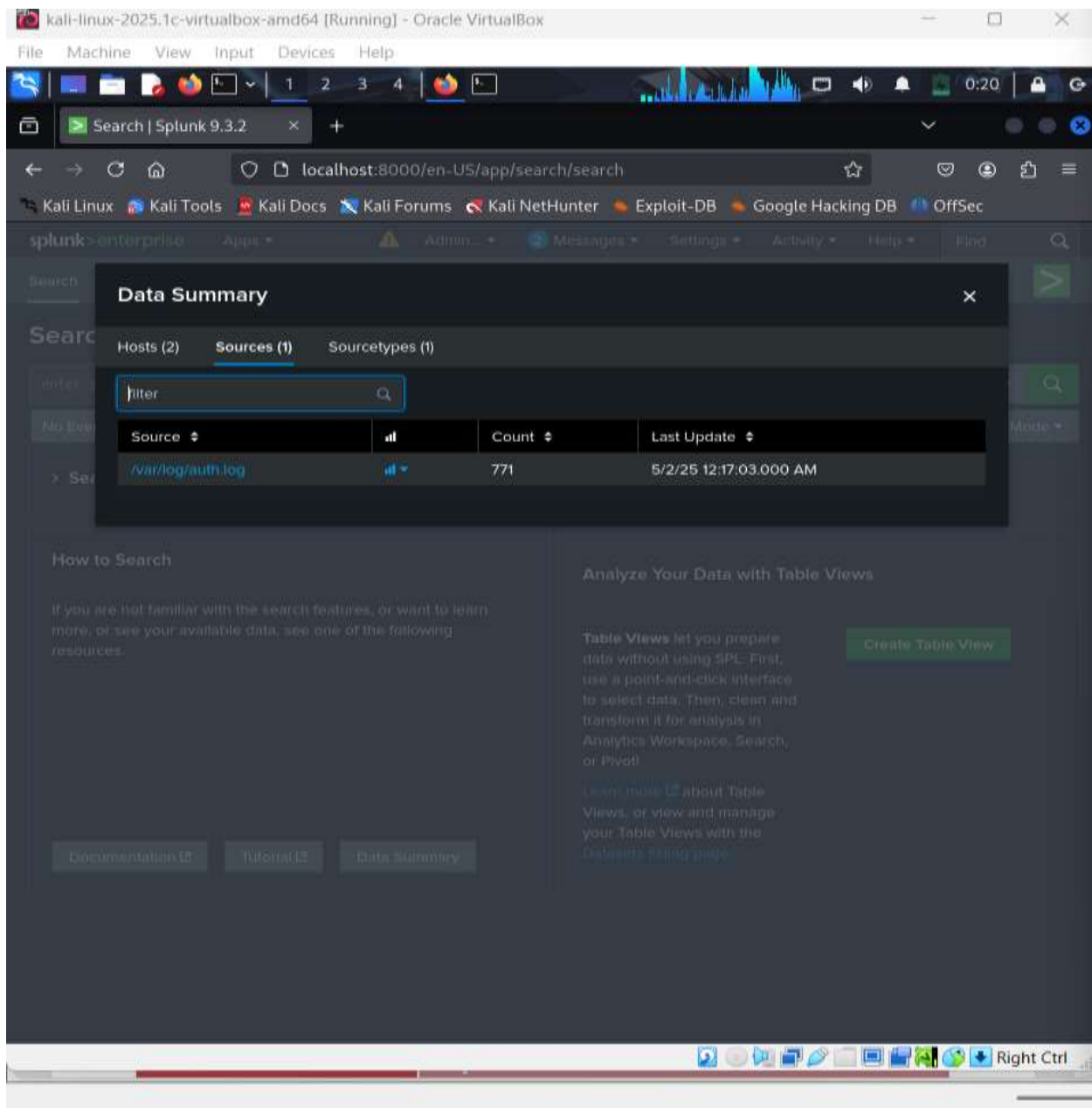
Phase 2: Successful Integration and Log Visualization via Splunk SIEM



```
Metasploitable3-ub1404 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConfig]/cliVerifyServerName for details.
Splunk username: admin
Password:
Added forwarding to: 192.168.56.102:9997.
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk /opt/splunkforwarder"
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConfig]/cliVerifyServerName for details.
Added monitor of '/var/log/auth.log'.
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk enable bootstrap-script
Error calling execve(): No such file or directory
Error launching command: No such file or directory
System start/stop links for /etc/init.d/splunk already exist.
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk list forwards-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk /opt/splunkforwarder"
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConfig]/cliVerifyServerName for details.
Active forwards:
    192.168.56.102:9997
Configured but inactive forwards:
    None
vagrant@metasploitable3-ub1404:~$
```





To accomplish Phase 2 of the ICS344 course project, we focused on setting up and validating a full SIEM (Security Information and Event Management) pipeline using Splunk to collect, forward, and visualize logs from a victim machine (Metasploitable3) into the centralized Splunk server running on a Kali Linux instance. The primary objective was to capture system activity logs from the victim and display them within Splunk's web interface for visual analysis. The process began by preparing the Splunk Enterprise (SIEM server) on Kali Linux, where we downloaded and installed the Splunk .deb package (splunk-9.3.2) using `wget` and `dpkg`, followed by the initial license acceptance and user credential setup. We then confirmed successful startup and ensured port 8000 (used by the Splunk web interface) was accessible. Once the server setup was complete, we shifted focus to the Splunk Universal Forwarder setup on the victim (Metasploitable3) machine. We downloaded a compatible .deb package of the forwarder (splunkforwarder-9.0.0) and installed it successfully, making sure the architecture matched the VM (amd64). After accepting the license and enabling it to run on system boot, we used the command line to forward logs to the server's IP address and port 9997 (`add forward-server 192.168.56.102:9997`). Credentials were verified and data forwarding was confirmed using `splunk list forward-server`, showing an active link to the SIEM. To collect meaningful data, we configured the forwarder to monitor `/var/log/auth.log`, which contains authentication attempts and is essential for identifying brute-force attacks or suspicious login activity. On the Splunk web interface, we accessed the "Search & Reporting" app, opened the Data Summary, and confirmed both the host (metasploitable3-ub1404) and the log file (`/var/log/auth.log`) were being correctly received and indexed. The number of indexed events (771 logs at the time) served as strong evidence of successful and continuous log transmission. This integration completes the critical component of Phase 2: enabling centralized monitoring, setting the stage for further forensic and security analysis. The accomplishment ensures that any real or simulated attacks on the victim environment can now be visualized, queried, and compared from a central dashboard, fulfilling the SIEM setup, log ingestion, and verification deliverables.